

LeetCode 2447

Number of Subarrays with GCD equal to K

Google Interview Question

Brute-Force Idea

Time Complexity: $O(n^2 \log A)$

Where 'A' is the biggest number in the array.

```
for (int i = 0; i < n; i++) {
    int g = 0;
    for (int j = i; j < n; j++) {
        g = __gcd(g, nums[j]);
        if (g == k)
            ans++;
    }
}
```

Observations for Optimal Approach

- **GCD never increases** when you extend a window. It either stays the same or it decreases.
- Suppose the current GCD is g . Then any future GCD is always a divisor of the previous GCD.
- Hence, the future GCD must divide g .
- So, if $g \% k \neq 0$, then any future GCD is a divisor of g . This guarantees that $\text{future_gcd} \% k \neq 0$.

Hence, k can never be a future GCD if the current GCD is not divisible by k .

Optimized Solution

```
class Solution {
public:
    int subarrayGCD(vector<int>& nums, int k) {
        int n = nums.size();
        int ans = 0;

        for(int i = 0; i < n; i++) { // Preserved from user input
            int gcd = 0;
            for(int j = i; j < n; j++) {
                gcd = __gcd(gcd, nums[j]);
                if(gcd == k) ans++;

                if(gcd % k != 0) break;
            }
        }
        return ans;
    }
};
```